

JOBA, Oman

WMO: 412870

Lat: 20.8681N

Lon: 58.2475E

Elev: 34

StdP: 100.92

Time Zone: 4.00 (E04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	14.0	-3.9	2.7	27.4	-1.4	3.4	27.9	11.0	21.4	9.5	23.1	2.6	350	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.8	42.6	21.8	40.8	22.3	39.2	22.9	27.4	33.6	27.0	32.7	26.6	32.3	6.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	29.1	25.8	21.2	29.1	25.3	20.6	29.1	87.4	33.5	85.7	32.8	84.0	32.3	32.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
12.2	11.2	10.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			4.2	27.4	2.6	0.8	2.3	28.0	0.8	28.4	-0.6	28.9	-2.5	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	21.0	22.6	26.1	29.9	33.0	33.0	30.5	29.4	29.3	28.8	25.7	22.2
	DBStd	4.29	1.63	2.07	2.03	2.04	1.64	2.15	1.91	1.53	1.39	1.46	1.66	1.75
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	2	1	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6438	341	352	498	596	712	689	636	601	579	584	471	379
	CDD18.3	3398	84	119	239	346	453	439	377	342	329	326	221	121
	CDH23.3	39548	672	1036	2485	4267	6382	6192	4462	3625	3529	3707	2189	1003
Wind	WSAvg	4.6	3.4	3.8	4.1	5.2	5.7	5.8	6.4	6.1	5.2	3.6	3.0	3.4
	PrecAvg	57	3	3	6	4	0	16	4	1	0	4	12	3
	PrecMax	193	58	21	64	76	1	136	62	12	0	44	98	16
	PrecMin	7	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	49	11	6	14	14	0	32	14	2	0	9	22	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	33.7	38.2	41.4	44.5	46.1	42.9	39.1	39.5	39.5	35.7	32.0
		MCWB	17.1	18.0	18.1	19.7	20.3	21.0	24.6	25.3	23.2	19.9	19.6	18.4
	2%	DB	28.9	31.4	35.9	39.9	42.3	43.2	39.1	37.2	37.1	37.8	34.1	30.3
		MCWB	17.6	18.1	18.3	19.6	21.4	21.9	25.2	25.1	23.6	20.3	20.2	18.5
	5%	DB	27.8	30.0	34.1	38.2	40.5	41.1	37.0	35.4	35.6	36.4	32.8	29.2
		MCWB	17.8	18.5	18.8	20.1	22.4	23.5	25.6	24.9	23.5	20.9	20.4	18.5
	10%	DB	26.9	28.6	32.3	36.4	38.8	38.7	35.1	33.9	33.9	34.8	31.5	28.1
		MCWB	17.4	18.5	19.0	20.4	23.6	24.5	25.4	24.9	23.9	20.9	20.7	18.4
	0.4%	WB	21.5	22.7	24.7	26.4	27.9	28.1	27.4	26.7	26.0	25.6	24.7	22.7
		MCDB	26.0	27.1	28.9	30.9	34.7	35.3	35.8	35.1	33.8	30.7	28.7	26.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	20.2	21.7	24.0	25.6	27.2	27.5	26.7	26.0	25.4	25.1	24.0	21.5
		MCDB	25.3	26.3	27.9	29.6	32.6	33.9	34.7	34.3	32.1	29.8	28.7	26.6
	5%	WB	19.6	21.0	23.2	25.1	26.8	27.1	26.2	25.5	25.1	24.6	23.2	20.6
		MCDB	25.0	25.9	27.6	29.1	31.7	32.9	33.6	33.0	31.1	29.3	28.2	26.0
	10%	WB	18.9	20.4	22.6	24.5	26.5	26.8	25.8	25.0	24.7	24.2	22.4	19.8
		MCDB	24.3	25.4	27.3	28.9	31.3	32.2	32.6	31.9	30.3	28.8	27.8	25.4
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.3	13.8	13.5	12.9	11.8	10.2	9.9	10.8	13.0	12.5	12.8
		MCDBR	13.7	14.6	16.1	15.8	15.2	15.9	13.2	12.4	13.1	15.4	13.9	13.8
		MCWBR	5.6	5.2	5.4	5.5	4.6	4.5	2.6	2.5	2.4	4.3	4.8	5.5
	5% WB	MCDBR	12.7	12.3	11.4	12.3	11.9	10.5	10.7	10.6	10.5	11.6	12.1	11.8
		MCWBR	5.7	4.4	3.6	3.9	2.8	2.3	2.5	2.6	2.2	2.9	4.6	5.2
Clear-Sky Solar Irradiance	taub		0.413	0.454	0.507	0.570	0.639	0.756	0.828	0.776	0.623	0.469	0.438	0.416
	taud		2.133	1.997	1.852	1.741	1.628	1.459	1.386	1.452	1.677	1.986	2.084	2.121
	Ebn at Noon		849	835	807	762	704	621	578	611	704	810	816	829
	Edh at Noon		143	173	208	236	262	307	331	311	245	173	149	140
All-Sky Solar Radiation	RadAvg		4.93	5.70	6.57	7.23	7.51	7.07	6.50	6.49	6.50	6.00	5.10	4.64
	RadStd		0.13	0.15	0.23	0.19	0.10	0.38	0.38	0.21	0.15	0.12	0.13	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	-0.63	-0.99	N/A	N/A	N/A

Nomenclature: See separate page